# Some Raindrops Can Slowly Wear Away Metal

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Raindrops may be hurting cars, bridges and statues more than people knew. But only some raindrops do it: the ones with an **electric charge**.

Drops can get a charge when they slide over leaves, glass or buildings. When these drops land on metal, they can pass the charge along. Over time, this makes the metal slowly wear away. This is called **corrosion**.

Scientist Zhongyuan Ni found this by accident. He left an **experiment** running while he went for coffee. When he came back, the **copper** looked different!

His team tested **copper** covered in Teflon, a strong **coating**. Charged drops wore through the **coating** and damaged the **copper**. Drops without a charge did no harm.

Ni says not to worry, because the effect is small. But after many years outside, it can add up. The discovery could help make better **coatings**.

## Word Treasure

**electric charge** -- An electric charge is a tiny bit of electricity that something can carry.

**corrosion** -- Corrosion is when metal slowly wears away, like rusting.

**experiment** -- An experiment is a test scientists do to learn something new.

**coating** -- A coating is a thin layer that covers and protects a surface.

**copper** -- Copper is a reddish metal used in wires and coins.

## Background you need

Metal can slowly wear away over time when it reacts with things like water and air.

People put coatings on metal, like paint or plastic layers, to protect it.

Rain falls on bridges, cars and statues every year, so even tiny effects can add up.

## Break it down

**WHO:** Scientist Zhongyuan Ni and his team

**WHAT:** They found that raindrops with an electric charge can slowly wear away metal, even through strong coatings

**WHERE:** On metal objects like cars, bridges and statues, tested in a lab

**WHY:** Drops pick up a charge when they slide over leaves, glass or buildings and pass it to metal

## Why it matters

Bridges, cars and playgrounds you use every day could last longer if scientists make better coatings.

## Test what you remember

**1. Which raindrops can hurt metal?**

- (A) All raindrops
- (B) Only snowflakes
- (C) Only warm drops
- (D) Drops with an electric charge

**2. How can drops get a charge?**

- (A) By sliding over leaves or glass
- (B) By falling at night
- (C) By being blue
- (D) By being cold

**3. What is it called when metal slowly wears away?**

- (A) Erosion of sand
- (B) Freezing
- (C) Melting
- (D) Corrosion

**4. What was Zhongyuan Ni doing when he made his discovery?**

- (A) Sleeping
- (B) Swimming
- (C) Getting coffee
- (D) Driving

**5. What happened to copper hit by drops with no charge?**

- (A) It melted
- (B) It turned gold
- (C) No harm was done
- (D) It broke in half

**6. What could this discovery help make?**

- (A) Better coatings
- (B) Faster cars
- (C) New coins
- (D) Bigger raindrops

Answer key (try the quiz first!): 1: D 2: A 3: D 4: C 5: C 6: A

# Questions to think about

**1. Zhongyuan Ni: Says not to worry because the effect is small, but it can add up over many years.**

MY THOUGHTS (at least 20 words):